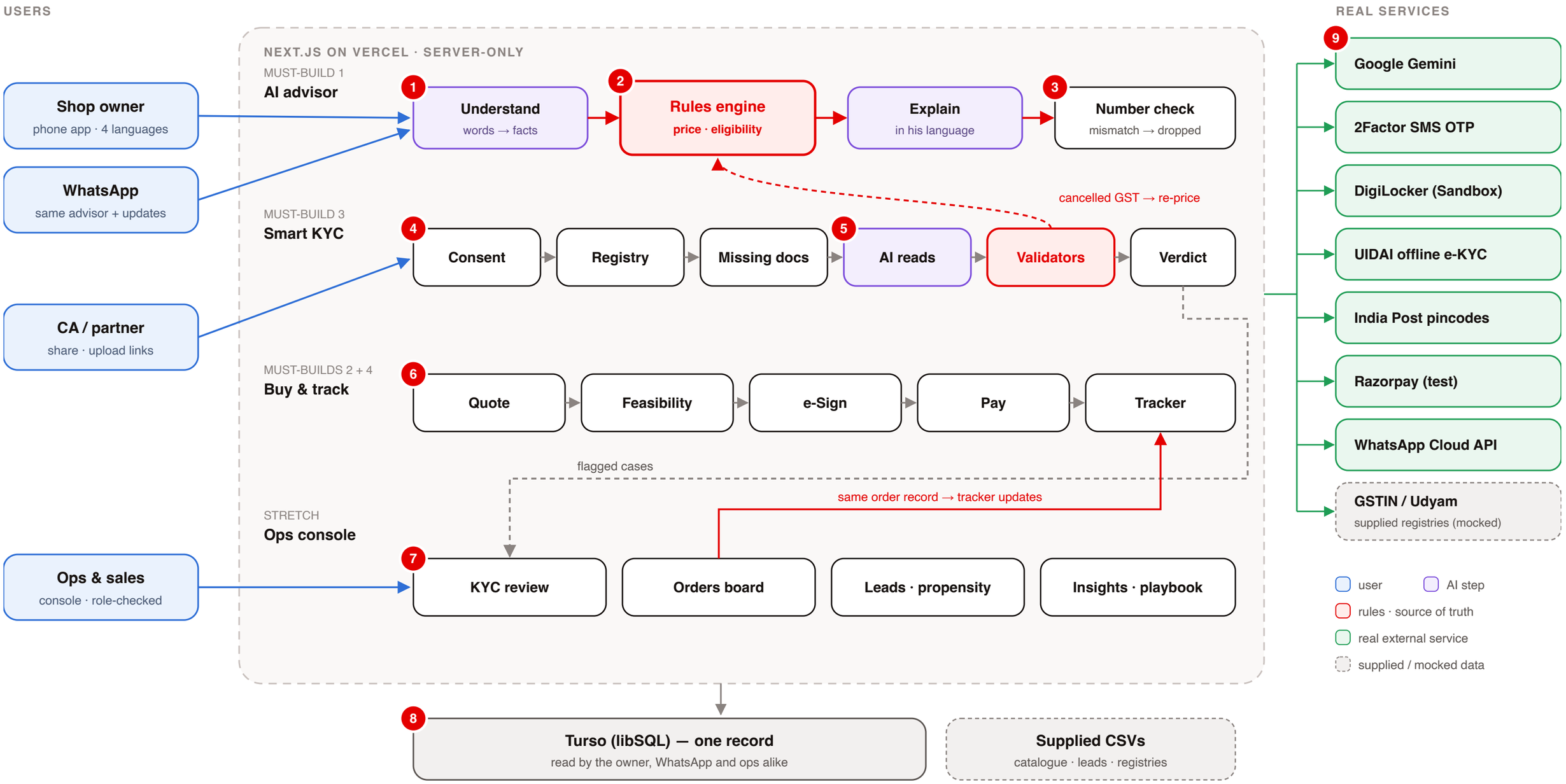
# Udyam Digital architecture — rules decide, AI explains

Price, eligibility and KYC verdicts come from tested code. The AI only understands words, reads pages and writes the explanation.




**Built in:** consent before KYC · masked IDs · role-checked ops + audit · signed webhooks · OTP limits · fail-closed services · one-tap export & delete

**One journey, end to end:** describe the shop (①) → engine picks the plan and price (②) → plan explained and number-checked (③) → consent and registry lookup (④) → only missing documents read and validated (⑤) → quote, sign, pay, track (⑥) → ops clears review and moves the order (⑦) — all on one record (⑧), using real services (⑨).

### 1 Understand — MUST-BUILD 1

The owner types or speaks in Hindi, Marathi, Tamil or English, on the app or WhatsApp. Gemini turns his words into structured facts — business type, staff, sites, city, GST — checked against a schema, and asks only for what is still missing.

### 2 Rules engine decides — SOURCE OF TRUTH

Eligibility (city tier, minimum staff and sites, a valid and active GSTIN, product dependencies) and the monthly price with GST come from the catalogue, in tested code, with a decision trace. The AI never sets a price or decides eligibility.

### 3 Explain, then check — ANTI-HALLUCINATION

Gemini words the plan in the owner's language using only the engine's result — why this plan, what would change it. Every ₹ and number in the reply is compared with the engine; any mismatch drops the reply. If the AI is down, the app says so instead of guessing.

### 4 Consent, registry first — MUST-BUILD 3 · DPDP

Consent is recorded per purpose before any document (marketing off by default). GSTIN checksum and status, and Udyam, are looked up first, so the owner uploads only what's missing — by upload, DigiLocker, Aadhaar offline e-KYC, or a link his CA opens without an account.

### 5 AI reads, rules judge — PLANTED PROBLEMS

Gemini vision reads each page in memory; the file is never stored. 20+ validators catch blurred scans, wrong GSTINs, bills older than 90 days, name mismatches and cancelled GST. Verdict: verified, re-upload with a fix, or human review. A cancelled GST re-prices the plan automatically.

### 6 Buy & track — MUST-BUILDS 2 + 4

A live quote with toggleable add-ons; India Post feasibility (no fibre → AirFiber); an order-form PDF signed with an SMS code bound to its SHA-256; Razorpay with a verified signature and one-tap retry; a tracker with a self-booked install slot and WhatsApp updates — no chase calls.

### 7 Ops closes the loop — STRETCH · BACK OFFICE

Reviewers clear flagged KYC cases with scores and reasons. Ops advance the same order record, so the customer's tracker and WhatsApp update at once. Leads carry a propensity score and a win-back list; insights and the segment playbook are computed from the 5,000 supplied leads.

### 8 One record, trust built in — DATA

Turso (libSQL) holds accounts, quotes, orders, extracted KYC fields, consent receipts and the audit log — one record read by the owner, WhatsApp and ops. IDs are masked, ops roles are checked, webhooks are signed, and the owner can export or delete his data in one tap.

### 9 Real services, fail-closed — INTEGRATIONS

Every external call runs server-side with keys in Vercel's environment: Gemini, 2Factor SMS, DigiLocker via Sandbox.co.in, UIDAI, India Post, Razorpay test mode, WhatsApp Cloud API. A missing key makes that feature refuse honestly. GSTIN/Udyam use the case's supplied registries, as briefed.

**STACK** Next.js 16 · React 19 · TypeScript · zod · Tailwind 4 · Drizzle ORM + Turso (libSQL) · pdf-lib e-sign · Vercel · Mumbai (bom1) · PWA, 4 languages

**VERIFIED BY** 270+ unit tests (Vitest) · 37 Playwright browser tests · KYC answer key K01–K12 · 5,000-lead backtest